

2(a)(i)	cartilage ; bod hyaline tissue	1
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Question	Answer	Marks
2(a)(ii)	<i>check 2(a)(i) if answer seems to be about, elastic tissue / smooth muscle / ciliated epithelium</i> I respiration <i>for breathing</i> any three from: C-shaped / incomplete rings / AW, in <u>trachea</u> A windpipe or irregular / plates , in, <u>bronchus / bronchi</u> ; bold blocks other structural feature ; e.g. surrounded by, perichondrium / layer of (dense) connective tissue two or three chondrocytes in, cavity / protective space / lacuna flexible ; <i>context is during, ventilation / breathing / changes in diameter of airway / bending of airway (but does not need stating)</i> supports / provides strength for, airway / trachea / bronchus ; <i>context is maintaining tubular structure I for structure</i> keeps airways open / prevents airways collapsing ; <i>(during, breathing / ventilation / changes in pressure)</i> <i>alternatives if incorrect gas exchange tissue stated in 2(a)(i):</i> allow 1 mark max ecf for feature / role (all information to be correct) e.g. elastic, tissue / fibres e.g. stretch / expand, and recoil R contract / relax smooth muscle tissue e.g. contracts and relaxes R stretch / recoil ciliated epithelium e.g. has ciliated epithelial cells and goblet cells / produces mucus and moves this by cilia out of airway R has mucous glands <i>allow 1 mark max if two different gas exchange tissues are described</i>	3

Question	Answer	Marks
2(b)	allow 'they' for 'telomeres' (as in the question text) ignore values for the number of bases 'lost' from telomere each replication as this varies any two from: ends of, telomeres / DNA / chromosomes, shorten / are lost / AW ; A telomeres shorten I telomeres lost context is each, DNA replication / (mitotic) cell cycle telomeres prevent the loss of, genes / genetic information ; ora presence of telomeres maintains, genes / genetic information I genetic material bod coding DNA ref. to continued / AW, mitosis / cell division / cell cycles, (of fibroblasts, needed for collagen production) ; bod <u>cell</u> replication AVP ; e.g. if telomeres lost, <u>ends</u> of chromosome may stick together (and affect mitosis) AW	2
2(c)	centrioles ; A centrosomes R if microtubules also stated	1

Question	Answer	Marks
2(d)	any two from: single DNA molecule ; A idea that the two joined sister chromatids are now separate entities (daughter chromosomes each with a DNA molecule) R chromatids, separated from their centromere I monovalent chromosome (tightly complexed with) histones or histone / basic, proteins ; (one) centromere ; I ref. to kinetochore / spindle microtubules correct ref. to state of condensation ; e.g. (early telophase) are, condensed / coiled / supercoiled (during telophase, begin to), decondense / uncoil / become diffuse R is decondensed, unless there is further explanation e.g. ‘...at the end of telophase’ / ‘becomes uncoiled and is decondensed’ <i>timing within telophase not needed but if stated reject a CON e.g. ‘decondense at the start of telophase’</i> AVP ; e.g. detached / detaching, from, spindle / spindle fibre microtubules / kinetochore, detaching / not attached / AW	2
2(e)(i)	endothelial (cell) ; R endothelium bod endothelium <u>cell</u> R squamous epithelial (cell)	1
2(e)(ii)	becomes thicker / more dense (ECM) / thicker layer ; I more collagen / collagen acts as a barrier I longer I increased concentration of components (<i>in Q text</i>) increased diffusion distance (for oxygen) ; decreased <u>rate</u> of diffusion (of oxygen) / less oxygen enters <u>per unit time</u> ;	2

Question	Answer	Marks
2(f)(i)	peptide (bond) ; R polypeptide bond bod amide (bond)	1
2(f)(ii)	<i>allow H bonds for hydrogen bonds</i> <i>any three from:</i> small amino acids / amino acids with small R-groups ; A glycine smallest amino acid A glycine / alanine, is a small amino acid glycine R-group H <u>and</u> alanine R-group CH₃ ; allows formation of triple helix ; A <i>ref. to</i> triple helical structure / three polypeptides wound into a helix / three helical polypeptide chains R <i>ref. to</i> triple helix forms a fibre R <i>ref. to</i> alpha helix allows tight, packing / coiling, of polypeptides AW or polypeptides held close together / AW / gly and ala on inside of coils ; <i>ref. to</i> hydrogen / H, bonds (involving, glycine / alanine) ; R strong H bonds R if covalent / disulfide / covalent cross-link bonds, stated <i>context is between polypeptides in the collagen molecule</i> <i>allow ecf if incorrect term used in mp3 or mp4</i> room / space, for, amine / amide / amino, groups to form hydrogen bonds ;	3